

X(3872) Three-Body Lineshape: Validation Note

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Summary

This note validates the three-body phase-space implementation used for the $\chi_{c1}(3872) \rightarrow D^0 \bar{D}^0 \pi^0$ and $\chi_{c1}(3872) \rightarrow D^0 \bar{D}^0 \gamma$ channels in the existing analysis and records the minimal extraction into a standalone package: `X3872ThreeBodyLineshape.jl`.

Implementation status

- Neutral-channel three-body phase space implemented via `xDDPhaseSpace`.
- Minimal API in new package: `rho_thr`, `lineshape`, `build_channels`.
- Charged-channel constants are placeholders and flagged for later validation.

Validation steps (commands + outputs)

- Ran the notebook to generate the phase-space plots:

```
/juliaup/bin/julia --project=.../X3872EffRange.jl -e 'include("notebooks/covariant_rho_thr"
```

- The plots were produced at:

- `X3872EffRange.jl/plots/phase_space_pi0-gamma.pdf`
- `X3872EffRange.jl/plots/pi0-gamma_ratio.pdf`

- Minor local patch applied for execution:

- add conversion for `WithThrE` to `rho_thr`
- guard a Pluto-only cell (`model_simpler`)

These are not committed to the source repository.

Plots

Delta from X3872EffRange

- Extracted a minimal object-based API: `X3872ThreeBody`, `denominator`, `lineshape`, `rho_thr`.
- Added tests and a parameter table with sources.
- Copied validation plots into the new repo for regression reference.

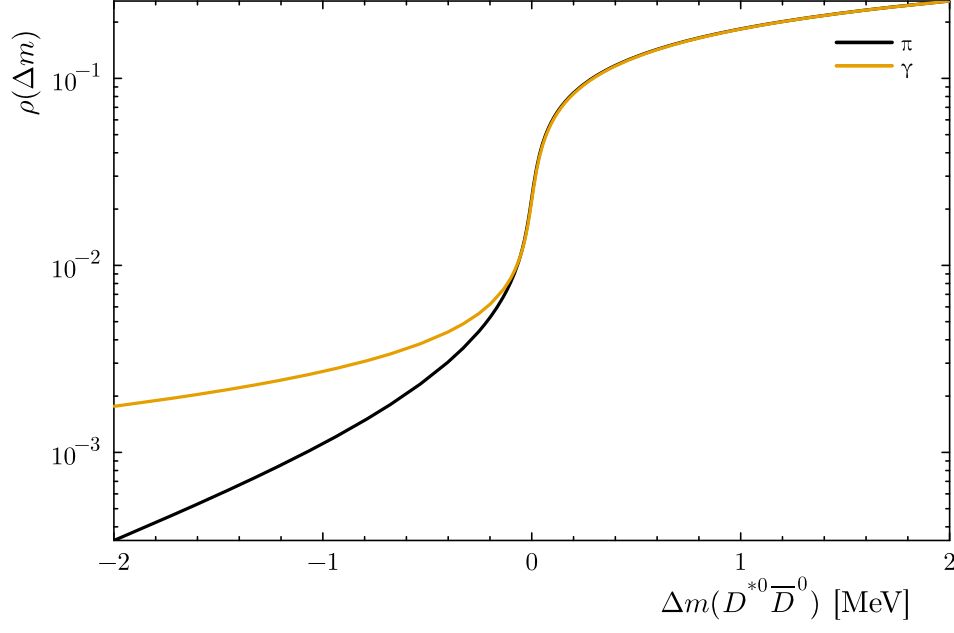


Figure 1: Energy-dependence of the phase-space factors of $\chi_{c1} \rightarrow D^0 \bar{D}^0 \pi^0$ and $\chi_{c1} \rightarrow D^0 \bar{D}^0 \gamma$, normalized at 30 MeV above threshold.

Next

Iteration 4 adds an analytic-continuation test and a minimal usage example. Iteration 5 updates this note with charged-channel validation once constants are fixed.

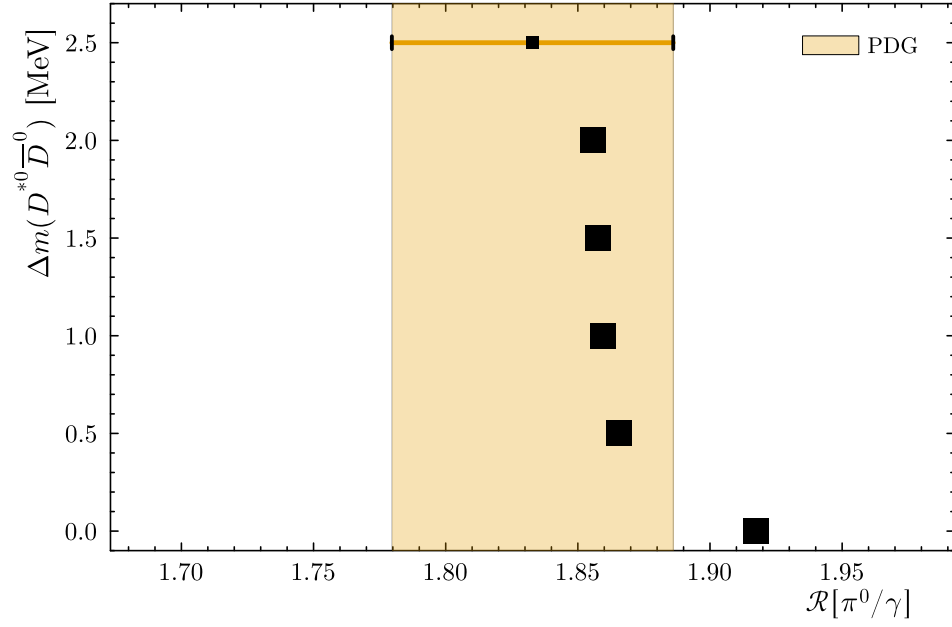


Figure 2: Ratio of the phase-space factors for $\chi_{c1} \rightarrow D^0 \bar{D}^0 \pi^0$ and $\chi_{c1} \rightarrow D^0 \bar{D}^0 \gamma$. The shaded band corresponds to the PDG branching ratio.